

# Joëlle Lafond, PhD

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## EDUCATION

|             |       |                                                                                                                                                                                                                                                                                                                                            |
|-------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018 – 2024 | Ph.D. | <b>Université de Montréal</b> , Département des Sciences Biologiques<br>Advisor: Bernard Angers<br><a href="#">Thesis</a> : Sexual and asexual reproduction in the fish <i>Chrosomus eos x eos-neogaeus</i> : context, mechanisms, and implications [in French]<br><i>Nominated for 2024 Best Thesis Award and the Rector's Honor List</i> |
| 2015 – 2018 | B.Sc. | <b>Université de Montréal</b> , Département des Sciences Biologiques<br>Program: Biological Sciences   Specialization: Biodiversity, Ecology and Evolution                                                                                                                                                                                 |

## APPOINTMENTS

|                       |                                                                                                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jan. 2026 – Current   | <b>FRQ Postdoctoral Fellow   University of Toronto</b> , Department of Ecology & Evolutionary Biology<br>Advisor: Stephen Wright (Dean of the Faculty of Arts & Science)                  |
| Jan. 2024 – Dec. 2025 | <b>FRQ Postdoctoral Fellow*   University of British Columbia</b> , Department of Zoology<br>Advisor: Judith E. Mank (Canada 150 Chair)<br><i>* Postdoctoral Researcher up to May 2025</i> |

## OTHER ACADEMIC EXPERIENCES

|             |                                                                                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018 – 2023 | <b>Teaching Assistant   Université de Montréal</b> , Département des Sciences Biologiques<br>Multiple contracts over the years for the following courses: <i>Evolutionary Genetics</i> , <i>Populations Genetics</i> , <i>Bioethics</i> , <i>Adaptative Ecology</i> . |
| 2020        | <b>Lecturer   Université de Montréal</b> , Département des Sciences Biologiques<br>Course: <i>Evolutionary Genetics</i> . Hybrid teaching due to COVID outbreak. Produced new teaching material and examinations, and improved old material.                          |
| 2020        | <b>Collaborative Researcher   McGill University</b> , Department of Biology<br>Anna L. Hargreaves' lab. Collected seed predation data in urban vs natural habitats.                                                                                                   |
| 2019        | <b>Research Intern   McGill University</b> , Department of Biology<br>Teruko Taketo's Lab. One month internship in cytometry and cell imaging.                                                                                                                        |
| 2017 – 2018 | <b>Research Assistant   Université de Montréal</b> , Département des Sciences Biologiques<br>Bernard Angers' Lab. Two summer contracts in genetics, epigenetics, population genetics (8 months).                                                                      |

## PUBLICATIONS

### PREPRINTS

10. **Lafond J.**, Saravanakumar S., Neiman M., Zufall R.A., Wright S.I. and Azevedo R.B.R. (2026). Reproductive mode and the evolution of mutation rate.  
[Preprint](#)

### ACCEPTED, IN PRESS & PUBLISHED

9. Sherin L., Johnson, B., **Lafond J.**, van der Bijl W. and Mank, J. (2026). From sexual conflict to sex-specific adaptation. *Philosophical Transactions of the Royal Society B (in press)*.  
[Accepted author copy](#)
8. **Lafond J.**, Lin Y., Fong L.J.M. and Mank J. (2026). Structural and Functional Divergence of the *Poecilia picta* Sex Chromosomes. *Molecular Ecology*.  
[Article](#)
7. Burkiewicz J., Carvalho J., Caporgno S., Devaux C., **Lafond J.**, Normandin E. and Joly S. (2026). Urbanization correlates with genetic and plastic variation of *Impatiens capensis* flower morphology. *Evolution Letters*, qrag009.  
[Article](#)
6. **Lafond J.**, Leung C. and Angers B. (2024). Asexuality shapes traits in a hybrid fish. *Nature Communications*, 2024(15): 7642. *Nominated for Science's Editor's Choice column*.  
[Article](#)
5. Hargreaves A. L., Ensing J., Rahn O., Oliveira F. M. P., Burkiewicz J., **Lafond J.**, Haeussler S., Byerley-Best M. B., Lazda K., Slinn H. L., Martin E., Carlson M. L., Sformo T. L., Dawson-Glass E., Chiufo M. C., Yalma L. Vargas-Rodriguez Y. L., Carlos I. García-Jiménez C. I., Gomes I. J. M. T., Klemet-N'Guessan S., Paolucci L., Joly S., Mehltreter K., Muñoz J., Buono C., Brodie J. F., Rodriguez-Campbell A., Veen T., Freeman B., Lee-Yaw J. A., Muñoz J. C., Paquette A., Butler J. and Suaréz E. (2024). Latitudinal gradients in seed predation persist in urbanized environments. *Nature Ecology and Evolution*, 2024(8): 1897–1906.  
[Article](#)
4. **Lafond J.** and Angers B. (2024). Maternal ploidy shapes reproductive pathways in the triploid hybrid *Chrosomus eos* × *eos-neogaeus*. *Molecular Ecology*, 33(5): e17264.  
[Article](#)
3. L'heureux E., **Lafond J.** and Angers B. (2023). First record of the invasive slug *Arion vulgaris* Moquin-Tandon, 1885 (*Gastropoda*, *Stylommatophora*, *Arionidae*) in Quebec (Canada). *Bioinvasions Records*, 12(1): 136–150.  
[Article](#)
2. Monette K., Leung C., **Lafond J.**, Wittische J. and Angers B. (2020). Inferring responses to climate warming from latitudinal pattern of clonal hybridization. *Ecology and Evolution*, 10(1): 307-319.  
[Article](#)
1. **Lafond J.**, Hénault P., Leung C. and Angers B. (2019). Unexpected Oogenic Pathways for the Triploid Fish *Chrosomus eos-neogaeus*. *Journal of Heredity*, 110(3): 370–377.  
[Article](#)

## FUNDING & FELLOWSHIPS

|             |                                                                                       |           |
|-------------|---------------------------------------------------------------------------------------|-----------|
| 2025 – 2027 | FRQ Postdoctoral Training Scholarship Program (Nature & Technologies section)         | 110 000\$ |
| 2023        | Excellence Scholarship for Ph.D. completed in < 5 years   Faculty of Graduate Studies | 12 000\$  |
| 2020        | Scholarship of Excellence   Hydro-Québec                                              | 60 000\$  |
| 2018        | Ph.D. Fast Track Scholarship   Faculty of Graduate Studies                            | 30 000\$  |
| Total       |                                                                                       | 212 000\$ |

## HONORS, AWARDS AND PRIZES

|             |                                                                         |          |
|-------------|-------------------------------------------------------------------------|----------|
| 2023        | Dissemination of Graduate Student Knowledge Program   FAECUM            | 500\$    |
| 2023        | International Travel Fund   FRQNT-GRIL                                  | 1000\$   |
| 2023        | 2 <sup>nd</sup> Prize at the Department's symposium (Oral presentation) | 150\$    |
| 2023        | QCBS Excellence Award   Quebec Center for Biodiversity Science          | 1350\$   |
| 2022        | Scholarship of Excellence   Faculty of Graduate Studies                 | 2 000\$  |
| 2021        | QCBS Fund Scholarship   Quebec Center for Biodiversity Science          | 3 372\$  |
| 2021        | 1 <sup>st</sup> Prize at the Department's symposium (Oral presentation) | 100\$    |
| 2021        | 1 <sup>st</sup> Prize at the Department's symposium (Speed-Talk)        | 100\$    |
| 2020        | Scholarship of Excellence   Fédération des caisses Desjardins du Québec | 1 500\$  |
| 2020        | Dissemination of Research Results   Faculty of Graduate Studies         | 1 514\$  |
| 2019        | Scholarship of Excellence   Étienne Magnin Foundation                   | 5 000\$  |
| 2019        | Travel Grant   The American Genetic Association                         | 200\$    |
| 2019        | 1 <sup>st</sup> Prize at the Department's symposium (Oral presentation) | 60\$     |
| 2018 – 2022 | Ichthyology Research Scholarship   Joseph Arthur Paulus Foundation      | 3200\$   |
| 2018        | Scholarship of Excellence   Departmental Funding                        | 1 500\$  |
| Total       |                                                                         | 21 546\$ |

## PRESENTATIONS

### INVITED TALKS

|      |                                                                                                                            |                       |
|------|----------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 2026 | <b>Guest lecture:</b> <i>Étudier la reproduction avec des modèles uniques: approches moléculaires et bio-informatiques</i> | Université de Moncton |
| 2024 | <b>Departmental Seminar:</b> <i>Hybridation et Reproductions Atypiques : Enjeux Écologiques et Évolutifs</i>               | Université de Moncton |

### INVITED SYMPOSIUM PRESENTATIONS

|      |                                                                                                                            |                                     |
|------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 2024 | <b>Talk:</b> <i>Maternal effects and shape change in a sexual/asexual fish complex</i> (Symposium on Adaptive Epigenetics) | <b>Evolution</b>   Montreal, Canada |
| 2024 | <b>Talk:</b> <i>Canceled – no funding</i> (Symposium on Hybridization and Unisexual Vertebrates)                           | <b>Evolution</b>   Montreal, Canada |
| 2024 | <b>Talk:</b> <i>Sexualization of the <i>Poecilia picta</i> X chromosome</i> (Symposium on Sex Chromosomes Evolution)       | <b>CSEE</b>   Vancouver, Canada     |
| 2023 | <b>Talk:</b> <i>Canceled – no funding</i> (Symposium on Hybridization and Unisexual Vertebrates)                           | <b>Evolution</b>   Albuquerque, USA |

## CONTRIBUTED PRESENTATIONS (12 talks, 2 speed-talks, 2 posters)

|      |                                                                                                                                           |                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 2026 | <b>Talk:</b> <i>Swift functional and structural evolution of sex chromosomes in Poecilia picta</i>                                        | <b>Evolution</b>   Cleveland, USA                                                 |
| 2026 | <b>Talk:</b> <i>Structural and Functional Divergence of Sex Chromosomes in Poecilia picta</i>                                             | <b>CSEE</b>   Toronto, Canada                                                     |
| 2025 | <b>Talk:</b> <i>Extensive Y Chromosome Degeneration and X Gene Migration in Poecilia picta</i>                                            | <b>CSEE</b>   Sherbrooke, Canada                                                  |
| 2024 | <b>Poster:</b> <i>Poecilia picta X chromosome sexualization</i>                                                                           | <b>Evolution</b>   Montreal, Canada                                               |
| 2023 | <b>Talk:</b> <i>Morphometry And Gametogenesis in a Fish Hybrid: The Role of Maternal Effects</i>                                          | <b>Evolution</b>   Albuquerque, USA                                               |
| 2023 | <b>Talk:</b> <i>Diploid or Triploid: How Mothers' Ploidy Affects Offspring Reproduction</i>                                               | <b>Departmental Symposium</b>   Montreal, Canada                                  |
| 2022 | <b>Talk:</b> <i>Morphometry And Gametogenesis in The All-Female Chrosomus eos-neogaeus: The Role of Maternal Effects</i>                  | <b>Evolution</b>   Online                                                         |
| 2022 | <b>Talk:</b> <i>Influence de la reproduction sur la morphologie dans un complexe d'hybrides vertébrés</i>                                 | <b>Departmental Symposium</b>   Montreal, Canada                                  |
| 2022 | <b>Speed-Talk:</b> <i>Influence de la reproduction sur la morphologie dans un complexe d'hybrides vertébrés</i>                           | <b>Departmental Symposium</b>   Montreal, Canada                                  |
| 2021 | <b>Talk:</b> <i>Sex determination in a unisexual vertebrate (Symposium on Genomic signatures and consequences of sex and asexuality)</i>  | <b>ESEB Satellite Symposium</b>   Online                                          |
| 2021 | <b>Speed-Talk:</b> <i>Détermination sexuelle chez les hybrides : un nouveau rôle pour les spermatozoïdes?</i>                             | <b>Interuniversity Research Group in Limnology</b>   Online                       |
| 2021 | <b>Talk:</b> <i>La place du déterminisme sexuel chez une espèce unisexuée</i>                                                             | <b>Departmental Symposium</b>   Montreal, Canada                                  |
| 2021 | <b>Speed-Talk:</b> <i>La place du déterminisme sexuel chez une espèce unisexuée</i>                                                       | <b>Departmental Symposium</b>   Montreal, Canada                                  |
| 2019 | <b>Talk:</b> <i>The Remarkable Reproductive Diversity of a Complex of Vertebrate Hybrids</i>                                              | <b>Quebec Center for Biodiversity Science</b>   Montreal, Canada                  |
| 2019 | <b>Poster:</b> <i>Quand la génétique ne dicte pas le phénotype : La reproduction particulière de petits poissons triploïdes</i>           | <b>Interuniversity Research Group in Limnology</b>   Orford, Canada               |
| 2019 | <b>Talk:</b> <i>Des reproductions différentes pour des mères identiques : Preuve que la génétique ne fait pas toujours le phénotype !</i> | <b>Departmental Symposium</b>   Montreal, Canada                                  |
| 2019 | <b>Talk:</b> <i>The Unexpected Diversity of Two Genetically Similar Hybrid Females in Their Reproductive Pathways</i>                     | <b>7th International Workshop on the Biology of Fish Gametes</b>   Rennes, France |

## ACADEMIC SERVICES

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### GRANTS & AWARDS JUDGING

2024            **National Grant** | Czech Science Foundation

### MANUSCRIPT PEER REVIEW

2026            **Journal of Evolutionary Biology**

2025            **Sexual Development**

2025            **Journal of Heredity**

2025            **Scientific Reports**

2025            **Advanced Science**

2019            **Ecology of Freshwater Fishes**

### PUBLIC OUTREACH

2022            **Researchers' Night (Montreal, Botanical Garden)**

Designed and led an interactive demonstration table to communicate my research to a broad public audience during an event attended by over 700 visitors.

2021            **Researchers' Night (Montreal, Biodome)**

Engaged in one-on-one discussions with visitors of all ages to demystify scientific research and the daily life of a graduate student. Facilitated small, highly interactive conversations throughout an event attended by over 300 participants.

### ORGANIZING COMMITTEE MEMBER

2021            **GRIL (Interuniversity Research Group in Limnology) Annual Symposium**

Giving help and direction on the creation of the Gather rooms for the Symposium.

2021            **Department's Annual Symposium**

Choosing the theme and guests, building the Gather platform, contacting potential guests and evaluators, building the website, and creating the event poster.

2020            **Departmental Research Day**

Building the event's page on Studium (Moodle) and producing the lab's graphic material for the event.

### ILLUSTRATIONS & VISUALIZATIONS (excluding authored paper)

2025            **Illustrations for Burkiewicz, J., Lavoie, J., and Joly, S. (Submitted to *Proceedings B*)**

Producing high quality visualization of the methodologies of the paper *Natural selection on *Impatiens capensis* flower morphology is consistent across urban and natural habitats.*

2021            **Seminar poster template of the Institut de Recherche en Biologie Végétale (Montreal)**

Producing high-quality template for the weekly seminars of the institute.